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Data Science Project

Introduction

Public trust in law enforcement is one of the most impactful factors of a successful and thriving society. A low or immeasurable level of confidence hinders the ability of effective cooperation, resolution, and safety. Attempting to understand the public trust in law enforcement through purely demographic factors is nearly impossible. In this analysis, over 40,000 observations are examined to understand that the public trust in law enforcement goes beyond the baseline of demographics and is more of a connected trust with institutional drivers such as the broader legal system and societal fairness. Both the courts and the police are responsible in keeping the rule of law fair and neutral, therefore going hand in hand when it comes to trust (Machura, et al., 2014). Examining how the intersection of various predictors affect trust in police will give us a clearer look in how to actually solve the issue of police confidence.

Methodology

The initial dataset had 56 variables with 50,116 observations, which was then reduced to one dependent variable and nine independent variables. Confidence in Police (1-10 scale) is the dependent variable in this analysis, which heavily influences the independent variables chosen to try and explain the variation and patterns in the data. The nine independent variables can be broken down into 3 smaller groups, for simplicity and order. The first group, demographics, consists of Age (Continuous), Gender (Male = 1, Female = 2), Years of Full-Time Education (Continuous. These three variables cannot only play a large role in an individual's trust of the police, but are crucial variables ensuring that the results found in other areas aren't just a reflection of the respondent's demographic profile. The second group of variables includes three institutional proxies: Confidence in the Courts and Legal System (0-10), Involvement in Political Affairs (1-4), and Placement on Political Spectrum (0-10 (0 = Far Left, 10 = Far Right)). By including these variables, trust in police specifically is isolated, rather than an overall trust in the government. These variables are somewhat similar but provide a unique insight into the trust of law enforcement, as they distinguish between institutional trust, engagement, and ideology. The third and final group consists of three experience-based variables: Background or Identity Leads to Unfair Societal Treatment (Yes = 1, No = 2), Someone in Your Household Being a Victim of Crime in Past Five Years (Yes = 1, No = 2), and National Belonging (0-10). This collection of variables will account for personal experiences when interpreting trust in police. Including these variables will tell stories about the responses that simple demographic or political variables won't, providing necessary context to our dependent variable.

Table 1: List of Variables and Measurements

Concept	R Variable Name	Scale / Measurement
Police Confidence	policetrust	Continuous (0 to 10)
Victim of Crime	victim	Binary (1 = Yes, 2 = No)
Unfair Treatment	background	Binary (1 = Yes, 2 = No)
National Belonging	countrybelong	Continuous (0 to 10)
Court Confidence	confCLS	Continuous (0 to 10)
Political Spectrum	polspec	Continuous (0 = Left, 10 = Right)
Political Affairs	polaffairs	Ordinal (1 = Close, 4 = Not at all)
Age	age	Continuous (Years)
Gender	gender	Binary (1 = Male, 2 = Female)
Education	educ	Continuous (Years of Full-Time)

After choosing the nine independent variables and the one dependent variable, the observations were reduced to 40,601 to ensure the integrity of the analysis. Many rows across the main variables contained N/A's, or missing values, resulting in a loss approximately 19% of the observations. The remaining sample is still extremely large, providing sufficient statistical power to detect the interactions and patterns in the data that are being searched for. The variables that provide the most missing data are 'Political Spectrum' with 7,489 (~15%) missing rows, and 'Confidence in Courts and Legal System' with 1,149 missing rows being the second most. All other variables saw under 1,000 missing rows each. Removing the missing values altogether felt reasonable rather than leaving out such important and descriptive variables, as political ideology specifically is known to be a significant predictor of institutional trust. Furthermore, with such a large sample size remaining after the removal of these null values, it felt safe to proceed. Preliminary checks suggest the missing data is random, meaning the exclusions did not significantly skew the representation presented in the data. Table 2 details the broad demographics found in the dataset, with a mean age of 51, quite an even split of gender, and 13 years of education on average. Other variables, such as identity and background leading to unfair treatment, have a mean of 1.9, showing the weight that a variable like that can have.

Figure 1, Distribution of Confidence in Police gives insight into the overall confidence in police throughout the sample. The distribution is skewed to the left, with a majority of the sample having a decent trust in the police. With this being a left skewed plot, the mean will sit below the median meaning the mode doesn't accurately represent the average. That being said, there are a couple of bins that are close to the mode, representing the overall high level of trust. There is still a decent amount of people on the lower end of the spectrum, as the second highest bin sits below average on the scale, as well as the 4 lowest bins having ~1,000-2,000 observations each.

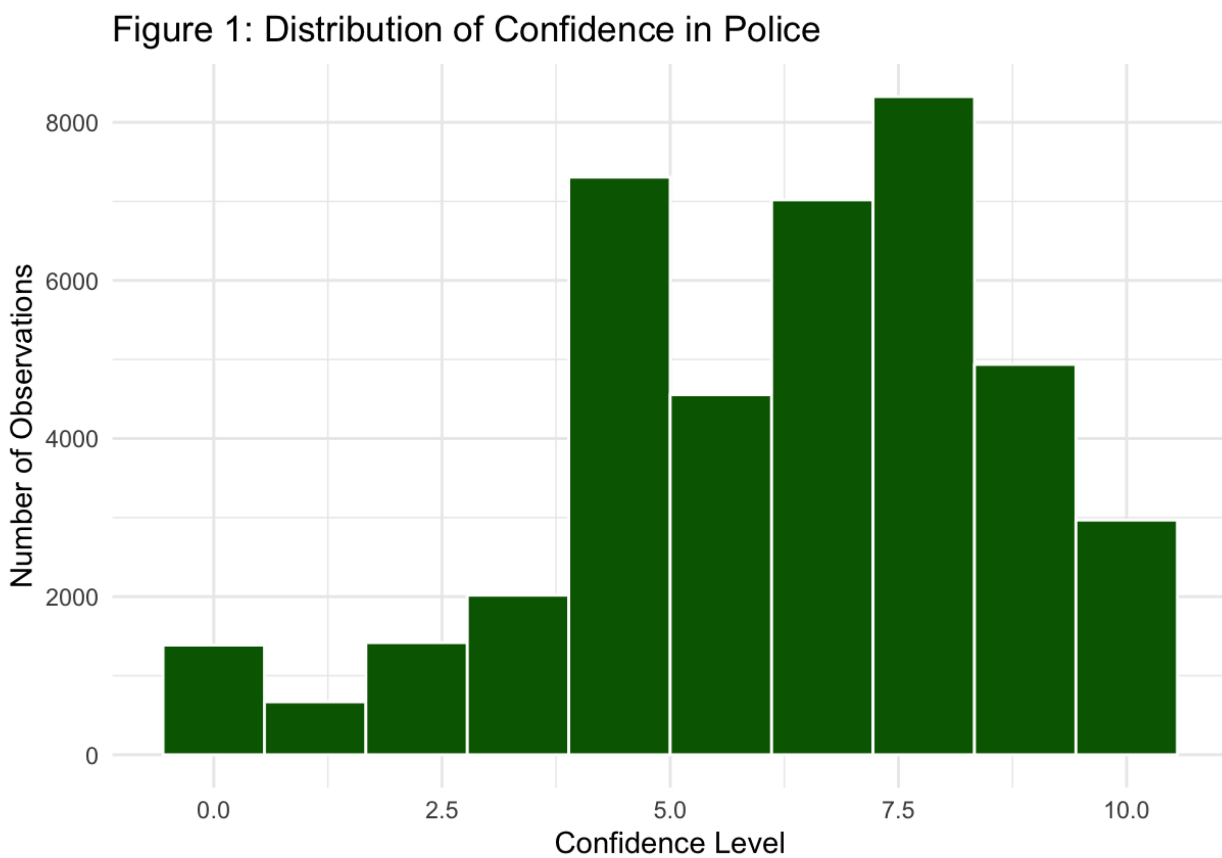

Figure 1: Distribution of Confidence in Police. Confidence Level on the X-axis and Number of Observation on the Y-axis, with a bin-width of 10 meaning each bin represents roughly one unit on the police confidence scale.

Table 2: Descriptive Statistics of the Analytical Sample

Statistic	N	Mean	St. Dev.	Min	Max
policetrust	40,601	6.426	2.465	0	10
age	40,601	51.733	18.317	15	90
gender	40,601	1.520	0.500	1	2
educ	40,601	13.504	4.020	0	50
countrybelong	40,601	7.913	2.174	0	10
confCLS	40,601	5.473	2.670	0	10
polspec	40,601	5.073	2.272	0	10
victim	40,601	1.891	0.312	1	2
polaffairs	40,601	2.550	0.913	1	4
background	40,601	1.914	0.280	1	2

Before running the multiple linear regressions, checking VIF values was a crucial step to make sure multicollinearity was low. VIF can be calculated with the following equation: 1 / (1 -

R^2). The general rule of thumb with VIF tests is you want the value for each variable to be roughly below 5. If the value is a considerate amount above 5, that means that it is too similar to another one of your variables and will inflate the numbers you find in the regression.

Once the dataset was in proper shape, the next step was to run multiple OLS regression models on a variation of the nine independent variables. The different groups that were mentioned earlier in the methodology were useful when running these variations, as pairing certain groups and variables with others gave insight to the true effect on police trust. The first of three multiple regressions included the three basic demographic variables, *Age*, *Gender*, and *Education*, as well as *National Belonging*. This regression gives a baseline on who the individuals are and how that effects their trust in law enforcement but still doesn't tell the whole story. The second regression includes all nine independent variables, giving lead to which ones are the strongest and have the biggest effect when zooming out. Figure 2 provides a visualization of all nine coefficients from the second model as well as their respective 95% confidence intervals, putting them in an ordered plot. The third and final regression includes an interaction term between *Victim of Crime* and *Unfair Treatment*, looking for how these two vulnerable variables affect one another. This last model included basic demographic variables as well in order to keep a baseline of individual profile. Figure 3 was created to demonstrate the effect that victimization has on those who have a perceived societal bias and those who don't. Visualizing the interaction term is important to fully understand the effect that these two variables have on each other and what differences in the two can have on the dependent variable.

Results & Analysis

Table 3: Demographic vs. Institutional Drivers of Police Trust

	Dependent variable:	
	Confidence in Police	
	Baseline	Institutional
	(1)	(2)
Age	0.006^{***}	0.004^{***}
	(0.001)	(0.001)
Gender	0.045^{*}	0.097^{***}
	(0.024)	(0.018)
Education (Years)	0.040^{***}	-0.011^{***}
	(0.003)	(0.002)
Unfair Treatment (Background)	1.052^{***}	0.468^{***}
	(0.043)	(0.032)
National Belonging		0.093^{***}
		(0.004)
Victim of Crime		-0.026

		(0.029)
Court Confidence		0.611^{***}
		(0.003)
Political Spectrum		0.066^{***}
		(0.004)
Political Affairs		0.027^{**}
		(0.010)
Constant	3.493^{***}	0.865^{***}
	(0.108)	(0.103)
Observations	40,601	40,601
R^2	0.020	0.472
Adjusted R^2	0.020	0.472
Residual Std. Error	2.441 (df = 40596)	1.792 (df = 40591)
F Statistic	204.375^{***} (df = 4; 40596)	$4,028.276^{***}$ (df = 9; 40591)
Note:		*p<0.1, **p<0.05, ***p<0.01

The following analysis utilizes multiple linear regression, allowing for the isolation and prediction of individual variables. Unless stated otherwise, all reported coefficients represent the effect of a one-unit change in the independent variable while holding all other variables constant. Before running the analysis VIF tests were run on each model in order to test for multicollinearity between the variables. Model 1's test returned all its variables below the threshold of 5, with no variable exceeding 1.05. This allows us to move forward with the regression without having to worry about certain variables jamming up the model.

The left side of Table 3 contains Model 1, a baseline demographic regression consisting of basic profile predictors, *Age, Gender, Education,* and *Unfair Treatment (Background).* Including these variables in a model by themselves allows us to see if 'who you are' determines trust in police. *Age* is the smallest driver here, with a coefficient of 0.006, meaning for every year increase in age, confidence in police increases by 0.006 on the confidence scale. This is pretty insignificant, proving that age doesn't play as big a role in trust as one might think. This predictor has a p-value under 0.01, meaning it is statistically significant, suggesting that older populations are slightly more inclined towards trust. *Gender* has a coefficient of 0.045, and with this being a binary variable (Male as the reference), we can interpret it as females being expected to have a 0.045-unit higher trust in police. While this estimate did not reach the $p < 0.05$ threshold, it still has a p-value under 0.1. This suggests a potential relationship that could use further investigation, and with such a big sample size tells us that the effect may even be smaller than our estimate tells us. Furthermore, the Standard Error for both the *Age* (0.001) and *Gender* (0.024) estimates are incredibly low, telling us that the relationship between these demographic predictors and trust is stable across the population, allowing us to trust this as a foundation for the model.

Education was also an impactful predictor in this initial model, with a coefficient of 0.040. This suggests that for additional year of education, confidence in the police increases by 0.040. The model is telling us that more education is associated with higher trust in the police, which could be correlated with increased institutional acceptance and support by those with advanced degrees. This can be related to the *Age* estimate being positive as well, indicating that older people are often more educated. *Education* is statistically significant under the $p < 0.01$ threshold, as well as obtaining a low Standard Error (0.003) meaning that we can deduce that this finding is not due to randomness and can be considered a stable indicator. *Unfair Treatment (Background)* is by far the most substantial predictor in this baseline model, with a coefficient of 1.052. This indicates a powerful positive relationship with confidence in the police, as respondents who believe they are treated fairly and don't perceive societal bias score more than a whole point higher on the 0-10 confidence scale than those who feel marginalized. The Standard Error of this variable remains low relative to its large coefficient, confirming that this is a wider issue than just the individuals surveyed. More so, this estimate also had a p-value under 0.01, proving to be statistically significant and not due to chance.

Model 1 has an R^2 value of 0.020, meaning that this demographic based model explains roughly 2% of the variance in trust. While this is a low value, it is still important information in our analysis, as it tells us that demographic indicators by themselves are not enough to determine trust in the police. Model 2 sets up more institutional based drivers that provide more explanation for what goes into that confidence. Furthermore, the F-Statistic (204.375) for Model 1 is not only high, but statistically significant, telling us that this model is more efficient and explanatory than a null model. The constant in Model 1 represents the predicted police trust score for a baseline respondent. In this case, this would be zero-year-old male, with zero years of education and feels as if they are treated unfairly by society due to their background or identity. With a value of 3.493, this tells us that there is a floor when it comes to police trust that exists.

On the right side of Table 3 is Model 2, or the institutional model, contains all nine of the predictors that were mentioned in the methodology. Before running this model, another VIF test was run to make sure these variables had low multicollinearity with each other, in which they did. None of the variables exceeded a VIF value of 1.16, proving the predictors to be independent of each other. There are some significant differences between this model and the previous one, all of which strengthen the predicting factor. To begin with, the differences in the predictors that we used in Model 1 highlight the importance of adding more than just demographic variables in this analysis. Figure 2 highlights that *Education* flipped signs completely, going from a coefficient of 0.040 to -0.011, serving as one of two negative estimates when all predictors are included. When the institutional trust variables are added, it is implied that more educated people are actually slightly less confident in the police, with one year of education resulting in a 0.011 decrease on the 0-10 confidence scale. With a Standard Error of 0.002 and a p-value below the 0.01 mark, we can assume that the negative relationship is not only statistically significant, but a solid foundation for the estimate outside of the survey.

Unfair Treatment's coefficient also saw a significant change, as it dropped from 1.052 to 0.093. As seen in Figure 2, this estimate sits in a separate tier with *Court Confidence,* serving as one of the most impactful. This is necessary to acknowledge, as adding the other predictors made it switch from having a full point effect on the police confidence scale to less than 0.5. Now, going from perceived societal bias to no perceived bias, only results in a 0.468 increase in trust, compared to it swinging a full point before. This suggests that some of the anger regarding unfair treatment is channeled through a general lack of confidence in the legal systems. With a low Standard Error (0.032) and a confirmed low p-value ($p < 0.01$), we can assume that this is not due to chance and is a reasonable predictor for the population. *Age* and *Gender* have relatively small effects compared to the other variables and didn't have a significant change in their coefficients. *Age* decreased from 0.006 to 0.004, remaining slightly positive but resulting in a minor decrease in trust from the previous model with each year added. *Gender's* coefficient practically doubled from 0.045 to 0.097, meaning with these institutional predictors, the switch from male to female leads to a 0.097 increase on dependent variable. Although doubling the size of the coefficient is significant, the relative increase is still minor due to the size of the original coefficient. Both of these demographic estimates have low p-values and Standard Errors, passing the threshold for statistical significance and low variability.

The predictors that were not in Model 1 are extremely important to this model and help paint a broader picture of what is related and affects police confidence. *Court Confidence* has the biggest coefficient in the whole model, serving as the strongest driver for the dependent variable. With a coefficient of 0.611, every point increase on the scale to measure confidence in the courts and legal system (0-10), results in a 0.611 increase on the police confidence scale. As seen in Figure 2 below, *Court Confidence* has by far the highest estimate, proving to be very influential in how citizens feel about law enforcement. This positive relationship means that those who hold more trust in the courts tend to feel similarly about the police. This makes sense intuitively, as these larger legal institutions tend to be closely related to law enforcement, connecting high trust in one to high trust in the other. This is an area of the government in which people feel strongly about, so this would most definitely be a place to target in policy reform. With a low p-value and Standard Error, this is very likely a statistically significant finding with low variance outside this model. Political engagement is represented by *Political Affairs* and *Political Spectrum* in this model, both of which have positive coefficients. *Political Affairs* measures how closely the respondent follows political affairs on a scale of 1 to 4, resulting in a coefficient of 0.027. Logically, this means that a one unit increase on this scale say, going from 1 – very closely, to 2 – closely, will result in the police confidence score increasing by 0.027. Those who follow political affairs very closely are associated with a lower police confidence score, which checks out. Police can often be seen in the wrong in political events and historically have been complained and protested about. Following these affairs closely will most likely expose you to more reasons to distrust the police. *Political Spectrum* measures where you would place yourself on the political spectrum on a scale of 1 (far left) to 10 (far right). With a coefficient of 0.066, a one unit increase on this political scale is associated with a 0.066-point increase in law enforcement confidence, proving to have a relatively small effect on police confidence. Based on this coefficient, we can assume the further right an individual is on the political spectrum, the more trust they would place in the police. While it is not a necessarily large estimate, it is still statistically significant while preserving a low Standard Error, therefore being useful to our model.

The final two variables in this model, *Victim of Crime* and *National Belonging* are important to analyze. As Figure 2 highlights, *Victim of Crime* is the lowest estimate in the model with a coefficient of -0.026, suggesting that switching from a victim of crime to a non-victim actually decreases the trust in police. While it is often assumed that experiencing a crime would shatter trust in police, the data tells a different story. This is most likely explained through victims having a sense of trust in the police post-crime, and an interaction with the police that most non-victims have most likely never experienced. Unfortunately, the Standard Error and p-value are relatively large, with this estimate not even reaching 90% confidence threshold. This suggests that the event of being a victim of a crime does not independently dictate trust levels and is mediated by some other variable. *National Belonging* acts as a positive indicator on police trust with an estimate of 0.093, suggesting that a one unit increase in national belonging (0-10) is related with a 0.093 increase on the police trust scale. Those who feel more comfortable and supported in their country are going to feel similarly to the authorities within. The p-value ($p < 0.01$) and Standard Error (0.004) indicate that police trust is partially rooted in a citizen's sense of belonging to the nation. We can assume the relationship is consistent across the dataset, and that individuals who feel a strong tie to their country are more likely to grant the police a higher baseline of trust.

The R^2 value for this second model jumps to 0.472 with these additional indicators, indicating that this make up of variables explains about 47% of the variance. This was expected after such a low value in Model 1 and suggests that institutional drivers play a much larger role in explaining police confidence than demographic predictors on their own. A value of 0.472 means there are most likely variables that could be added to further enhance the predicting power of this model, but this is still a very strong model with a wide range of variables. The F-Stat (4.028.276) confirms this, as it is not only much larger than Model 1's, but shows how much power these variables have compared to a model with none. Lastly, the constant here represents the value on the police scale if all of these variables were at their reference or zero. The constant drops lower than in Model 1, indicating that if a citizen has no trust in courts or institutional predictors, their trust in the police will be compromised as well.

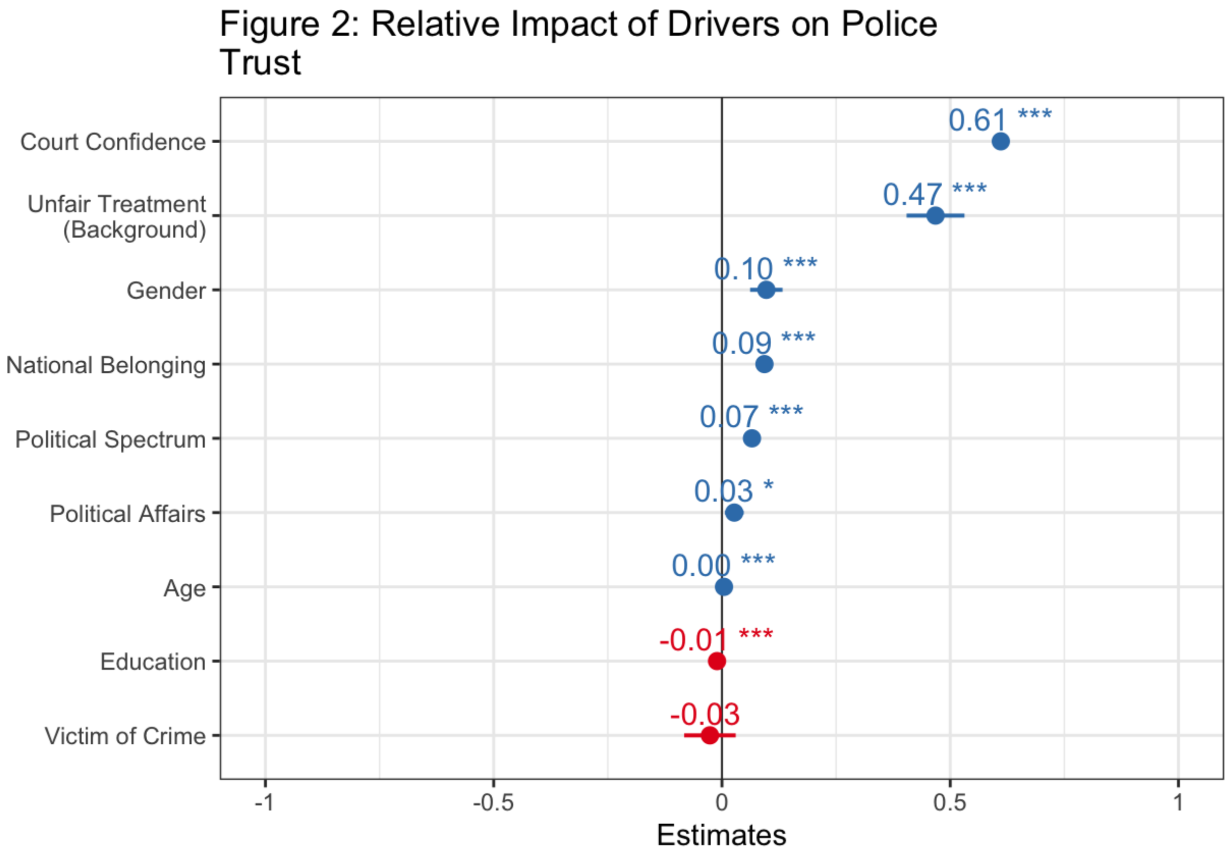

Figure 2: Relative Impact of Drivers on Police Trust. Each variables coefficient value is on the X-axis with the independent variable it corresponds to on the Y-axis. The vertical line at zero represents whether the estimate is negative or positive, while the whiskers on certain dots represents 95% confidence intervals. The stars next to each dot represents the significance level, using the same scale as the regression tables.

Table 4: Victimization and Identity Interaction Model

	Dependent variable:
	Confidence in Police
Victim of Crime	-0.334***
	(0.101)
Unfair Treatment	0.984***
	(0.049)
Age	0.006***
	(0.001)
Gender	0.044*
	(0.024)
Education	0.041***

	(0.003)
Victim * Background	0.306^{***}
	(0.110)
Constant	4.616^{***}
	(0.083)
Observations	40,601
R^2	0.020
Adjusted R^2	0.020
Residual Std. Error	2.440 (df = 40594)
F Statistic	138.182^{***} (df = 6; 40594)
Note:	*p**p***p<0.01

The third and final regression model consists of three of the basic demographic variables, *Age, Gender,* and *Education,* as well as *Victim of Crime, Unfair Treatment,* and an interaction term of the two, *Victim * Background.* Before running this model, a VIF test was performed on these variables and while the demographic predictors presented acceptable values (none larger than 1.05), the highest observed VIF in this model was 6.76 for *Victim* and 6.75 for the interaction term. While this does slightly exceed the threshold of 5, the rule of thumb is pretty loose and can be based on judgement, a 6.76 is perfectly fine especially due to the fact that it is an interaction term with built in correlation (Stack Exchange, 2013). With our VIFs being low, we can move on with certainty in our predictors.

The interaction term *Victim:Background* is the main point of this model and tells us a lot about how being a victim of a crime and perceiving societal bias can affect trust in police. With a coefficient of 0.306, we can interpret this as positive impact of fair treatment on police trust being significantly stronger for those who have also been a victim of a crime. This is evident in Figure 3, as the vertical gap in trust is much larger for victims than for non-victims when moving from unfair to fair treatment. Non-victims are a completely different story, as their trust levels increase gradually in response to unfair treatment. To quantify the visual in Figure 3, the baseline trust for a non-victim perceiving unfair treatment is 4.616. This interaction term is statistically significant ($p < 0.01$, SE = 0.110), confirming that one social's identity and trust with police is not uniform, but is impacted by whether they've experienced a crime. Social equity is also confirmed as a solid predictor of trust for the general population.

The coefficient of *Unfair Treatment,* 0.984, remains a significant driver in the interaction model. For the general population, moving from a perception of being treated unfairly to fairly results in a predicted increase of nearly a full point on the confidence scale. With a low Standard Error (0.049) and p-value ($p < 0.01$) we can assume this applies to the general population, proving that societal fairness is a foundation to trusting police. *Victim of Crime* has a coefficient of -0.334, representing the impact of victimization for someone who feels they are treated unfairly by society. The negative coefficient combined with a significant p-value suggests that those who feel marginalized and experience a crime creates the lowest baseline of trust. The

Standard Error (0.101) may seem high, but with this being a binary variable where one group is smaller than the other, there is going to be more uncertainty.

Age, Gender, and *Education* all have positive coefficients in relation to police trust. *Age* (0.006) continues to be a statistically significant estimate, indicating that a one-year increase in age is associated with a 0.006 increase on the police scale. *Gender* has a coefficient of 0.044 yet only has a p-value below 0.1, meaning once victimization and individual background are accounted for, whether you are male or female becomes a lot less powerful of a predictor. Similarly, *Education*'*s* estimate flips back to positive at 0.041, meaning a one-year increase in education is associated with 0.041 increase on the police confidence. When accounting for the interaction term, highly educated individuals are associated with higher police trust. This estimate has a p-value below 0.01 and has a low Standard Error, meaning the variability for this iteration of the variable is low, while remaining statistically significant.

The R^2 of this model drops back down to 0.020, the same as Model 1, once again meaning that this group of variables explains 2% of the total variance. While this is exceptionally low, the model is focused on the effects of identity and trauma on our dependent variable. A study with a low R^2 value can still provide deep insight into specific effects. In this case the interaction term of 0.306 identifies a critical point to focus on in terms of policy. An F-Statistic of 138.182 that is also statistically significant indicates that the model is sound and the predictors collectively offer an improvement over a null model. Model 3 has a constant of 4.616 meaning a victim of a crime who feels societal bias starts with a 4.616 on the police confidence scale. Once again, this proves that the baseline for police confidence is low but present even with this interaction term in the model.

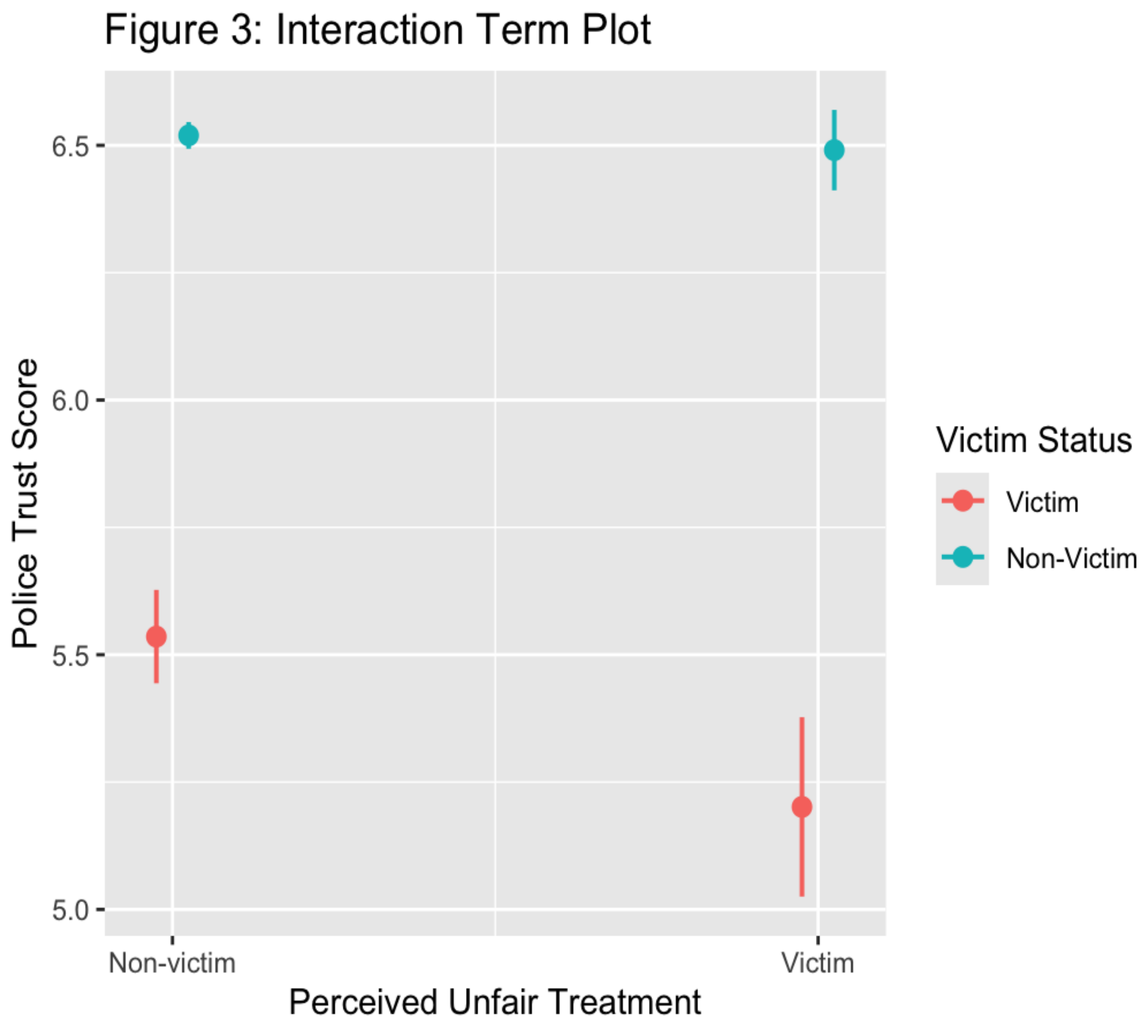

Figure 3: Interaction Term Slope Plot. This plot consists of Perceived Unfair Treatment on the X-axis and Police Confidence on the Y-axis, with red representing being a victim of a crime, and blue representing not being a victim. The whiskers on each dot represents respective 95% confidence intervals.

Discussion

The findings in the result section tell us that *Court Confidence* is the biggest driver in Model 2, having the most impact on police confidence when all variables are in play. This means that public trust is tied to the perception of the legal system as a whole, proving that a focus on improving on a broader scale is the best path forward. Changing the police system alone is not enough, as a corrupt or inefficient court system will make it harder even if police reform is done right. Transparency with the public is a place to start, possibly providing regular reports about the recent activity in courts. More so, fairness and responsiveness to all cultures, languages, and special populations is key, as if a group doesn't feel represented by the legal system there is no reason for them to trust them (NACM, n.d.). Another way to make the legal system feel more accessible and efficient would be to prioritize judge's responses and activity. As the leaders of the courtroom, ensuring that the courtroom is somewhere that citizens are not afraid of, but rather somewhere that they feel safe and interested will be better for all parties involved. It is also the job of everyone in the courtroom to always preserve and improve the law, as unequal treatment and a lack of promoting justice will only make this overarching issue worse (National Judicial College, 2013). Understanding that the courts and police are tied is a necessary step towards improving confidence in both sides of the legal system and needs to be a primary focus of reform.

A large focus of this analysis was how *Unfair Treatment (Background)* affected an individual's trust in the police. How people perceive they are treated by society based on their identity is almost as important as the legal system itself. This relates to the Procedural Justice Theory (PJT), claiming that when the police show integrity and fairness, they gain trust and cooperation (Tansill, 2024). Improving the quality of the interactions is a critical step to not only reduce the bias felt by many, but to simultaneously improve the standard and confidence of the police. PJT consists of four stages, the first being improving the style of interactions and the neutrality of decision making. The second stage plays off the first, stating that fair and neutral policing will result in a stronger bond between individual and authority. This also plays into *National Belonging* findings as social bonds between both parties will result in higher senses of belonging for citizens, not only improving police confidence, but overall comfort and trust in the state. Third, these bonds set a precedence for trust in citizens that will make them want to cooperate and allow the authorities to police in an appropriate manner. Finally, because of all these stages, a more sustainable mode of cooperation will present itself, as citizens won't feel a need to comply due to fear (Bradford, et al., 2023). In addition to court reform, PJT will make citizens more comfortable with police and lead to a desire to cooperate.

Another variable that stood out was *Victim of Crime,* proving that being a victim of a crime doesn't automatically destroy one's trust in the police. This suggests that the experience of the crime is less impactful than the response. Depending on how the police respond to the victim, they may actually feel more trust in the police if it was handled with efficiency and care. Furthermore, this finding connects to *Court Confidence* as well, implying that the manner in which the courts handle a victim's experience is just as important. A functional and fair court has proven to be a large driver in police trust, and this would only make victims feel more supported and cared for by the legal system. Other variables had low coefficients, including *Age* and *Education*, proving that trust in law enforcement isn't solely based on how old or educated an individual is. While older and more educated people tend to have more trust in the police, it is primarily driven by institutional and social fairness, which are things the government can actively change.

When running multiple linear regressions, there will be limitations that arise from your data collection and analysis. Firstly, a main limitation of all data analysis is proving causation, as you can never definitively state that one variable causes the other. For example, in this analysis you cannot definitively say that *Court Confidence* causes *Police Confidence.* This limitation introduces omitted variable bias, in which there is a possibility that other variables that were not included, such as crime rate in this example, influenced the relationships being analyzed and could further explain the variance. Furthermore, the data relies on self-reported data, such as

Unfair Treatment (Background) and has a broad definition that could be interpreted differently by each individual. Subjectivity can make reading these variables tricky, but it is a limitation that you have accept in analysis.

Conclusion

Ultimately, this analysis demonstrates that confidence in the police is not purely based on demographics, but rather a reflection of institutional trust and performance. Predictors such as *Court Confidence* and *Unfair Treatment (Background)* had significantly higher coefficients than other variables, emerging as the dominant indicators. It is clear that police confidence is anchored in the perceived integrity and fairness of the broader legal system and the equity in which it treats its citizens. The path to a more trusted police force does not lie directly within the police department. It requires zooming out and being transparent in all facets of the legal system and addressing systemic issues in order to create a two-sided relationship.

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AI Usage Statement

This report used Google Gemini (2026) as a technical assistant. The AI was used with these verbatim prompts:

- “how to check amount of NAs in R”
- “all of my variables are under 5 percent NA except one which is 14 percent, what should i dp about these?”
- “i ran drop_na() on my dataset and nothing changed”
- “how to set up histogram in ggplot”
- “how to set up boxplot in r”
- “is this useful? This doesnt feel like a great graph”
- “can you help me with mean confidence by gorup, i want to implement a facet wrap but i think theres too much data”
- “Error in `combine_vars()`:
  ! At least one layer must contain all faceting variables: `education`
  ✕ Plot is missing `education`
  ✕ Layer 1 is missing `education`
  Run `rlang::last_trace()` to see where the error occurred”
- “reg1 <- lm(policetrust ~ age + gender + educ + countrybelong + confCLS + polspec + victim + polaffairs + background, data=policeclean)
  reg1
  reg2 <- lm(policetrust ~ age + gender + educ + background, data = policeclean)
  reg2”
- “how to use stargazer to put them side by side”
- “for some reason it is not showing the beta for unfair treatment background for the baseline model, it is showing that beta under political affairs”
- “stargazer(reg2, reg1,
  type = "text",
  title = "Regression Results: Factors Influencing Confidence in Police",
  dep.var.labels = "Confidence in Police (v1)",
  column.labels = c("Baseline Model", "Full Institutional Model"),
  covariate.labels = c("Age", "Gender", "Education", "Unfair Treatment Background", "Country Belonging", "Confidence in Courts", "Political Spectrum", "Victim of Crime", "Political Affairs"),
  out = "regression_table.txt")
  this is my code and the output, andi keep getting an error but i dont know whats worng”
- “can i use stargazer on summary(), i want to present the data summary cleanly”
- “why is my interaction term different when i do victim*background in the regression and when i make a separate variable like victim_background <- victim * background, the first is -0.305, and the second is 0.26”

- “reg3 <- lm(policetrust ~ victim + background + victim_background + age + gender + educ, data = policeclean)
  reg3”
- “stargazer(reg3,
     type = "latex",
      title = "Regression on Baseline Factors including Victim/Background Interaction Term",
   covariate.labels = c("Victim",
                    "Background",
                    "Victim * Background"
                    "Age",
                    "Gender",
                    "Education"),
  dep.var.labels = "Confidence in Police",
  no.space = TRUE)
  does this look right, i also want to remove the variable victim_background”
- “can you help me with an interaction plot”
- “how to make a table for my variables in R”
- “best way to get summary statistics from R into word”
- “can you help me get all three rgressions in stargazer (reg1, reg2, reg3), i cant get them to line up correctly”
- “how to calculate slope of line in R”

Full R Code

```
library(tidyverse)
police <- readRDS("ECON30130data.rds")
view(police)

#Cleaning data
policeclean <- police %>% select(ends_with("v1"),
                          ends_with("v3"),
                          ends_with("v4"),
                          ends_with("v7"),
                          ends_with("v15"),
                          ends_with("v20"),
                          ends_with("v29"),
                          ends_with("v37"),
                          ends_with("v43"),
                          ends_with("v50"))
policeclean <- rename(policeclean,'policetrust' = 'v1', 'age' = "v3", 'gender' = "v4", 'educ' = "v7",
        "countrybelong" = 'v15',
```

```
                'confCLS' = 'v20', "polspec" = 'v29', 'victim' = 'v37', 'polaffairs' = 'v43',
       'background' = 'v50')
summary(policeclean$age)
summary(policeclean$gender)
summary(policeclean$educ)
summary(policeclean$countrybelong)
summary(policeclean$confCLS)
summary(policeclean$polspec)
summary(policeclean$victim)
summary(policeclean$polaffairs)
nrow(policeclean)
policeclean <- policeclean %>% drop_na()
view(policeclean)


#Model 1: Baseline Characteristics Regression on Police Confidence
reg1 <- lm(policetrust ~ age + gender + educ + background, data = policeclean)
reg1
library(stargazer)
stargazer(reg1,
      type = "text")
#Model 2: Full Institutional Regression on Police Confidence
reg2 <- lm(policetrust ~ age + gender + educ + background + confCLS + polspec + victim +
       polaffairs + countrybelong, data=policeclean)
reg2
stargazer(reg2,
      type = "text")

#Model 3: Baseline Model Including Victim/Background Interaction
policeclean <- policeclean %>%
  mutate(
    victim = factor(victim, levels = c(2,1),
            labels = c("Non-victim", "Victim")),
    background = factor(background, levels = c(1,2),
              labels = c("Unfair", "Fair"))
  )

reg3 <- lm(policetrust ~ victim * background + age + gender + educ, data = policeclean)
reg3
stargazer(reg3,
      type = "text",
      title = "Regression on Baseline Factors including Victim/Background Interaction Term",
      covariate.labels = c("Victim * Background",
                 "Age",
                 "Gender",
                 "Education"),
```

```
        dep.var.labels = "Confidence in Police",
        no.space = TRUE)

# Baseline Model (Exporting)
stargazer(reg1,
        type = "html",
        out = "Table_1_Baseline.html",
        title = "Table 1: Baseline Demographic Regression",
        covariate.labels = c("Age", "Gender (Female)", "Education (Years)", "Unfair Treatment
          (Background)"),
        dep.var.labels = "Confidence in Police",
        digits = 3)

# Full Institutional Model (Exporting)
stargazer(reg2,
        type = "html",
        out = "Table_2_FullModel.html",
        title = "Table 2: Full Institutional Regression Model",
        covariate.labels = c("Age", "Gender", "Education", "Unfair Treatment",
                        "Court Confidence", "Political Spectrum",
                        "Victim of Crime", "Political Affairs", "National Belonging"),
        dep.var.labels = "Confidence in Police",
        digits = 3)

# Interaction Model (Exporting)
stargazer(reg3,
        type = "html",
        out = "Table_3_Interaction.html",
        title = "Table 4: Victimization and Identity Interaction Model",
        covariate.labels = c("Victim of Crime", "Unfair Treatment", "Age",
                        "Gender", "Education", "Victim * Background"),
        dep.var.labels = "Confidence in Police",
        digits = 3)

#side by side table

names(coef(reg2))
names(coef(reg3))

main_labels <- c("Age",
            "Gender",
            "Education (Years)",
            "Unfair Treatment (Background)",
            "National Belonging",
            "Victim of Crime",
            "Court Confidence",
```

```
                "Political Spectrum",
                "Political Affairs"
                )

stargazer(reg1, reg2,
        type = "html",
        out = "Main_Regression_Table.html",
        title = "Table 3: Demographic vs. Institutional Drivers of Police Trust",
        column.labels = c("Baseline", "Institutional"),
        order = c("age", "gender", "educ", "background", "countrybelong",
                "victim", "confCLS", "polspec", "polaffairs"),
        covariate.labels = main_labels,
        intercept.bottom = TRUE,
        digits = 3)

#Plot 1: Distribution of Confidence in Police
ggplot(policeclean, aes(x = policetrust)) +
  geom_histogram(bins = 10, fill = 'darkgreen', color = 'white') +
  theme_minimal() +
  labs(title = "Figure 1: Distribution of Confidence in Police",
      x = "Confidence Level",
      y = "Number of Observations")

#Plot 2: Relative Drivers on Police Trust
plot_model(reg2,
          show.values = TRUE,
          value.offset = .3,
          sort.est = TRUE,
          axis.labels = c("Victim of Crime",
                        "Education",
                        "Age",
                        "Political Affairs",
                        "Political Spectrum",
                        "National Belonging",
                        "Gender",
                        "Unfair Treatment (Background)",
                        "Court Confidence"),
          title = "Figure 2: Relative Impact of Drivers on Police Trust",
          vline.color = "black") +
  theme_bw()

#Plot 3: Victim:Background Interaction Plot
install.packages("sjPlot")
library(sjPlot)
plot_model(reg3,
          type = "int",
```

```
          title = "Figure 3: Interaction Term Plot",
          axis.title = c("Perceived Unfair Treatment", "Police Trust Score"),
          legend.title = "Victim Status") +
  theme_bw()


#Table 2: Summary Statistics for Key Variables
summary_data <- policeclean %>%
  select(policetrust, age, gender, educ, countrybelong, confCLS, polspec, victim, polaffairs,
          background)
stargazer(as.data.frame(summary_data),
          type = "text",
          title = "Table 1: Summary Statistics for Key Variables",
          digits = 2,
          covariate.labels = c("Police Trust (0-10)", "Age", "Gender", "Education", "Country
            Belonging",
                          "Confidence in Courts", "Political Spectrum", "Victim of Crime", "Political
            Affairs", "Unfair Treatment - Background"))

#summary stats table
stargazer(policeclean,
          type = "html",
          out = "summary_stats.html",
          title = "Table 2: Descriptive Statistics of the Analytical Sample",
          digits = 3,
          summary.stat = c("n", "mean", "sd", "min", "max"))

#VIF Checks
library(car)
vif(reg1)
vif(reg2)
vif(reg3)

#Table 1: Variable table and measurements
library(tidyverse)
library(knitr)
var_table <- tribble(
  ~"Concept",             ~"R Variable Name", ~"Scale / Measurement",
  "Police Confidence",    "policetrust",      "Continuous (0 to 10)",
  "Victim of Crime",       "victim",        "Binary (1 = Yes, 2 = No)",
  "Unfair Treatment",     "background",       "Binary (1 = Yes, 2 = No)",
  "National Belonging",   "belonging",        "Continuous (0 to 10)",
  "Court Confidence",     "courts",           "Continuous (0 to 10)",
  "Political Spectrum",   "polspec",        "Continuous (0 = Left, 10 = Right)",
  "Political Affairs",   "polaffairs",     "Ordinal (1 = Close, 4 = Not at all)",
  "Age",                 "age",             "Continuous (Years)",
```

```
  "Gender",              "gender",          "Binary (1 = Male, 2 = Female)",
  "Education",           "educ",            "Continuous (Years of Full-Time)"
)
#variables table
write.csv(var_table, "variable_table.csv", row.names = FALSE)
```